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Epidemiological and digital syndromic surveillance data on dengue, chikungunya, and SARI in Brazil

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Vector-borne diseases, such as dengue and chikungunya, along with air-borne diseases like influenza and COVID-19, are prone to epidemics, which increases the demand for real-time outbreak monitoring. Developing such systems requires harmonized datasets for calibration and validation. In this study, we created a dataset containing official disease notifications for dengue, chikungunya, and severe acute respiratory infections (SARI) across Brazilian states for over a decade. The dataset integrates Google Trends search data for each disease and all associated symptoms, organized by the corresponding epidemiological week. By providing this integrated resource, we aim to encourage the use of alternative online data to explore associations between digital search behavior and official disease incidence, thereby supporting the development of nowcasting models for early outbreak detection to inform timely public health responses and decision-making.

Background & Summary

Epidemic-prone diseases such as arboviruses (e.g. dengue and chikungunya) and severe acute respiratory infections (SARI), which includes influenza and COVID-19, pose significant risks to public health and impose a substantial burden on health systems worldwide[1–6]. The direct public health expenditure for managing dengue fever alone during the 2012/2013 epidemic in Brazil was estimated to be between $468 million and $1.2 billion[7]. Beyond the direct economic costs, long-term conditions[8,9] can further impact the burden for many years after an outbreak.

The rapid shift from endemic patterns to epidemic outbreaks places added strain on health systems[10–14]. Severe cases can result in hospitalizations or even death, particularly among socioeconomically vulnerable populations and high-risk groups like children, immunocompromised individuals, people with comorbidities, and the elderly[15–17]. Predicting the next outbreak is complex and involves interactions among climatic, biological, social, and human behavioral factors[18,19]. Nevertheless, early detection of outbreaks is critical, enabling authorities to prepare and mitigate the impact, while allowing the public to adopt preventive measures[20].

In response to these threats, traditional surveillance systems often face challenges due to delays in reporting and disseminating information[21]. To mitigate this issue, researchers have increasingly turned to alternative data sources, with internet search queries showing notable promise for early detection of disease trends[22,23]. In particular, platforms such as Google Trends can capture real-time search behavior related to diseases and symptoms, offering timely insights that enhance and complement traditional surveillance efforts.

To support syndromic surveillance efforts, here we present a dataset that integrates official disease notifications with real-time behavioral indicators, specifically internet search trends related to arboviral and respiratory diseases in Brazil. These diseases present common symptoms, particularly during the initial days of infection, like sudden fever and generalized pain. More specific symptoms, such as joint pain (chikungunya), rash (dengue), and respiratory distress (influenza and COVID-19), typically emerge later, though there is considerable variability in both the timing and severity of symptoms[24]. By harmonizing disease notifications with

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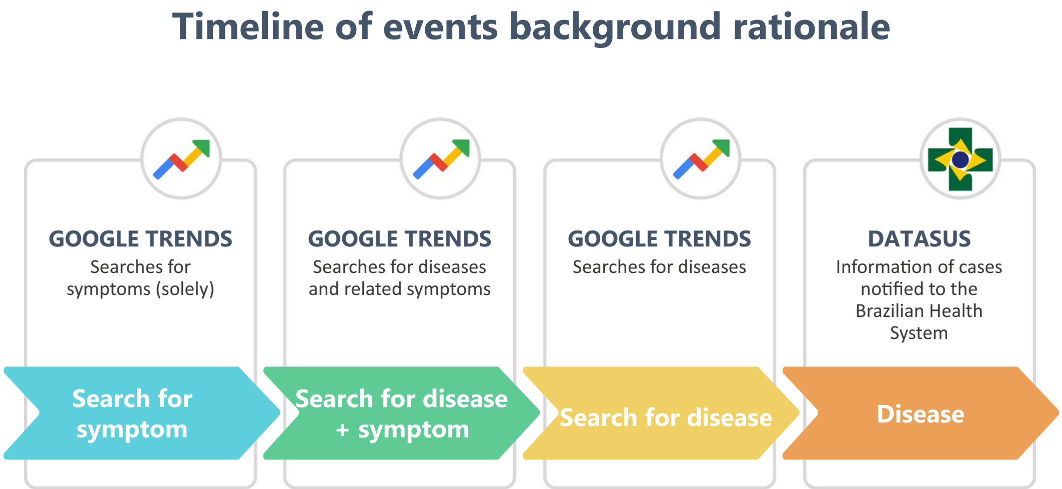

Fig. 1 The hypothetical sequence of events leading to disease notification, from initial symptom searches to official case reporting in health surveillance systems such as DATASUS.

symptom-related online search behavior, the dataset enables the development and evaluation of surveillance tools suited for complex epidemiological contexts.

The dataset was organized based on the hypothesis that online search behavior follows a sequence of events leading up to the official reporting of cases. As shown in Fig. 1, we propose that individuals first search for symptoms, as they may be unsure of what they have. The first searches typically combine the disease name with symptoms, which typically occur when the disease starts to appear in the media and in their local communities. Next, individuals search for the disease itself, indicating greater certainty about their condition. Finally, this progression is eventually followed by the official reporting of cases in health surveillance systems, such as DATASUS. This sequence suggests that monitoring symptom-related search trends on Google Trends could act as an early indicator of disease outbreaks, potentially preceding official case notifications and providing valuable insights for real-time epidemiological surveillance[25].

Methods

Study site. Brazil is a continental-sized country with 203.1 million inhabitants, according to the 2022 census[26]. The country has 27 federative units (26 states and one federal district) with diverse population distribution, varying income levels, differing human development indices, and contrasting prevalence rates of arbovirus and respiratory diseases. Historically, arboviral diseases have been more common in tropical and equatorial regions, while respiratory diseases[27] are more prevalent in the subtropical region[28]. Brazil's climate is generally categorized into equatorial, tropical, semi-arid, highland tropical, and subtropical zones[29], each influencing disease dynamics. The northern equatorial region, marked by high temperatures and humidity, creates ideal conditions for mosquito breeding, which contributes to the ongoing transmission of arboviruses like dengue and chikungunya[30]. Semi-arid areas in the northeast, characterized by extended droughts, may limit mosquito proliferation but still face arboviral outbreaks during the wetter seasons, especially in populated areas[31]. Recent temperature increases caused by climate change have been favoring the emergence of arboviral outbreaks in southern regions of Brazil[32]. Additionally, in regions with seasonal temperature variations, characterized by colder and drier months, there is a periodic increase in respiratory infections, including influenza or SARI caused by other viruses[27,33–35].

Data sources. Our dataset integrates diseases and associated symptoms data from official public sources with Google search results, organized by the corresponding epidemiological week. Both data sources and data extraction methods are detailed below.

Official public sources (diseases and symptoms). Dengue and chikungunya cases were obtained directly from the Brazilian Notifiable Diseases Information System (Sistema de Informação de Agravos de Notificação, SINAN, ftp://ftp.datasus.gov.br/dissemin/publicos/SINAN/), Brazil's main platform for recording and monitoring cases of notifiable diseases. It employs standardized forms for each disease that collect data related to patient identification and the primary characteristics of the illness. Established in the 1990s, SINAN was created as part of a government initiative to improve public health surveillance. In 1998, it became mandatory for municipalities and states to report notifiable diseases across the country[36]. Over the years, SINAN has broadened its scope to encompass a broader range of diseases, reflecting changing public health priorities and the rise of new epidemics. Dengue cases have been reported to SINAN since 2001[37], while chikungunya was included in 2017[38].

SINAN data is publicly accessible through the file transfer protocol (FTP) service maintained by Brazil's Unified Health System data platform (DATASUS). Dengue notifications are stored in the SINAN-Dengue database, while chikungunya notifications are available in the SINAN-CHIK database. Data extraction was performed using the R package *microdatasus*[39], which facilitates access to SINAN records.

Variable	Year of Availability		Description
	Dengue	Chikungunya	
DT_DIGITA*	2010	2021	Date of first notification entry into the system
DT_NOTIFIC	2010	2015	Date of notification form completion
DT_SIN_PRI	2010	2015	Date of symptom onset
SG_UF_NOT	2010	2015	Federative unit of notification
CLASSI_FIN	2010	2015	Final classification
FEBRE	2013	2017	Symptom: Fever
MIALGIA	2013	2017	Symptom: Myalgia
CEFALEIA	2013	2017	Symptom: Headache
EXANTEMA	2013	2017	Symptom: Rash
VOMITO	2013	2017	Symptom: Vomiting
NAUSEA	2013	2017	Symptom: Nausea
DOR_COSTAS	2013	2017	Symptom: Back pain
CONJUNCTIVITIS	2013	2017	Symptom: Conjunctivitis
ARTRITE	2013	2017	Symptom: Arthritis
ARTRALGIA	2013	2017	Symptom: Arthralgia
PETEQUIA_N	2013	2017	Symptom: Petechiae
LEUCOPENIA	2013	2017	Symptom: Leukopenia
LACO	2013	2017	Test: Tourniquet test
DOR_RETRO	2013	2017	Symptom: Retro-orbital pain

Table 1. Description of raw variables extracted from the SINAN database for notifications of dengue and chikungunya, including associated symptoms and the initial year of data availability. *Missing variable in the databases for the years 2014 to 2020.

In the SINAN system, all suspected cases are notified according to the case definition for suspected cases established by the Pan American Health Organization (PAHO)[40], which is based on clinical presentation. In our database, we included cases classified as probable, plausible, or confirmed, which encompass both laboratory-confirmed cases and those confirmed through clinical and epidemiological criteria. Discarded cases were excluded from the dataset. Case definitions for dengue and chikungunya follow PAHO guidelines. For dengue, symptoms should include a fever lasting 2 to 7 days, accompanied by at least two of the following: nausea, vomiting, rash, myalgia, headache, retro-orbital pain, petechiae, a positive tourniquet test, or leukopenia. For chikungunya, a suspected case is defined by the sudden onset of fever and severe arthritis or arthralgia that cannot be attributed to other conditions. For both diseases, it is required that individuals have either lived in or traveled to endemic or epidemic regions within 14 days prior to symptom onset, or who have an epidemiological link to a confirmed imported case[41]. Since the SINAN notification includes data on symptoms as defined in the notification form, these reported symptoms were also extracted. The variables selected from the SINAN database are detailed in Table 1.

Data on Severe Acute Respiratory Infections (SARI)—caused by influenza, COVID-19, or other viruses—were obtained from the Influenza Epidemiological Surveillance Information System (SIVEP-Gripe) and downloaded directly from OpenDataSUS[42], the Brazilian Ministry of Health's open data portal. The respective datasets can be located by entering 'SRAG' in the portal's search field. Unlike the reporting of arboviruses to SINAN, only hospitalized SARI cases are reported to this system.

SIVEP-Gripe is Brazil's primary surveillance system for SARI cases. Established in 2000, it began reporting SARI cases in 2009 as part of the response to the H1N1 influenza pandemic. In 2020, in response to the COVID-19 pandemic, the system was further adapted to monitor the concurrent circulation of SARS-CoV-2, influenza, and other respiratory viruses[43]. Since then, the mandatory notification of all suspected or confirmed COVID-19 cases that meet the criteria for SARI has been incorporated into SIVEP-Gripe, ensuring comprehensive tracking of severe respiratory infections during public health emergencies.

Severe acute respiratory infection (SARI) is defined as an individual with ILI who presents with dyspnea, respiratory distress, persistent chest pressure, oxygen saturation $\leq$94% in room air, or bluish discoloration of the lips or face[44]. ILI is characterized by fever accompanied by cough or sore throat, with symptom onset within the last seven days[45]. In contrast, under the framework of universal COVID-19 surveillance, ILI involves an acute respiratory condition defined by at least two of the following signs and symptoms: fever, chills, sore throat, headache, cough, runny nose, or disturbances in smell and taste senses. In children, nasal congestion is also considered in the absence of another specific diagnosis. In elderly individuals, additional criteria for severity should be considered, such as syncope, mental confusion, excessive drowsiness, irritability, and loss of appetite[43]. For suspected COVID-19 cases, fever may be absent, and gastrointestinal symptoms (such as diarrhea) may be present.

The SIVEP-Gripe dataset has been available through Opendatasus since 2009, which defines the period covered in our database. Similar to the SINAN system, a predefined list of reported symptoms is entered during the notification process and made available; therefore, we included them in our data extraction. The selected variables, along with the year they were added to the database, are detailed in Table 2. In 2019, variables for

Variable	Year of Availability	Description
DT_DIGITA	2010	Date of first notification entry into the system
DT_NOTIFIC	2010	Date of notification form completion
DT_SIN_PRI	2010	Date of symptom onset
SG_UF_NOT	2010	Federative unit of notification
CLASSI_FIN	2010	Final classification
FEBRE	2010	Symptom: Fever
TOSSE	2010	Symptom: Cough
GARGANTA	2010	Symptom: Sore throat
DISPNEIA	2010	Symptom: Dyspnea
DESC_RESP	2010	Symptom: Respiratory distress
DIARREIA	2010	Symptom: Diarrhea
VOMITO	2019	Symptom: Vomiting
PERD_OLFT	2020	Symptom: Loss of smell
PERD_PALA	2020	Symptom: Loss of taste
DOR_ABD	2020	Symptom: Abdominal pain
FADIGA	2020	Symptom: Fatigue
POS_PCRFLU	2019	Positive PCR for influenza
PCR_SARS2	2020	PCR result for SARS-CoV-2
AN_SARS2	2020	Antigen result for SARS-CoV-2
POS_AN_FLU	2020	Positive antigen for influenza

Table 2. Raw variables extracted from the SIVEP-Gripe database for analysis include the year of availability, a brief description, and the corresponding codes along with their meanings.

"vomiting" and "positive PCR for influenza" were added. Beginning in 2020, fields related to COVID-19 symptoms, PCR and antigen tests for COVID-19, and antigen tests for influenza were incorporated.

Since SIVEP-Gripe records only severe cases of respiratory diseases that require hospitalization, milder cases are reported in another system, e-SUS Notifica. However, the absence of mandatory reporting and challenges related to data cleaning and standardization on this database makes it highly susceptible to notification bias and underreporting. As a result, e-SUS Notifica is not suitable for accurately tracking the total number of influenza cases in the country. In contrast, the mandatory reporting of severe cases makes SIVEP-Gripe the most reliable system for monitoring respiratory disease cases in Brazil.

Web Search Data (diseases and symptoms). Data on public interest in diseases and symptoms was obtained from Google Trends through its API. We extracted the Google Trends search index, Related Topics and Related Queries. Data extraction was performed using the gtrendsAPI package in R[46], which facilitates direct access to Google Trends data.

The Google Trends search index measures the relative popularity of a search term over time and a geographical region. It ranges from 0 to 100, with 100 representing peak interest during the specified period and 0 indicating minimal or no search activity. This index does not reflect absolute search volumes; rather, it compares the search interest for a given term to its peak within the selected timeframe. The search index is normalized to account for variations in total search volumes across different regions and time periods. It is calculated by dividing the number of searches for a specific term by the total number of Google searches in the same location and timeframe. This normalization ensures that the index reflects relative interest, preventing larger regions or those with higher search volumes from skewing the data. As a result, it allows for meaningful comparisons across geographic areas and time periods, emphasizing fluctuations in public interest instead of absolute search metrics numbers.

Characteristic symptoms for chikungunya, dengue, COVID-19, influenza, and Severe Acute Respiratory Infections (SARI) were identified based on a consultation of official documents and scientific publications, including guidelines from the Brazilian Ministry of Health[45,47], the World Health Organization (WHO)[48,49], the Pan American Health Organization (PAHO)[50], and selected peer-reviewed articles[51]. Symptom selection was guided by clinical distinctiveness and reporting frequency, prioritizing those most indicative of each disease. The most characteristic symptoms and their associated diseases are presented in Table 3.

To broaden the scope of search behavior analysis and account for local linguistic diversity, we also compiled a list of alternative expressions used to describe each symptom, including popular and regional terms commonly used throughout Brazil. Additionally, Google Trends distinguishes between searches for specific "terms" and broader "topics". A search by "term" retrieves data based on exact keywords, while a "topic" search encompasses related terms and variations, including misspellings and regional differences. After identifying relevant search topics, we used their corresponding Freebase ID codes from Wikibase whenever possible, to include variations in spelling and language. These unique identifiers help disambiguate search terms, ensuring that queries remain focused on the intended diseases and symptoms while excluding unrelated meanings. For instance, the topic "Fever" (symptom) has the Freebase ID "*/m/01s08g*", distinguishing it from non-clinical uses, such as in music or entertainment. A complete list of topics, search terms, Freebase ID codes, and their variations used in this study is presented at the Table 4.

Symptom	Chikungunya	Dengue	COVID-19	Influenza	SARI
Arthralgia	✓				
Chills			✓		✓
Cough				✓	
Dyspnea			✓		
Fatigue		✓	✓	✓	
Fever	✓	✓	✓	✓	✓
Headache		✓			
Alteration of taste or smell			✓		
Loss of taste or smell			✓		
Muscle stiffness					✓
Myalgia		✓		✓	
Nausea		✓			
Retro-orbital pain		✓			
Sore throat			✓	✓	
Vomiting		✓			

Table 3. Most characteristic symptoms of each disease, used for identifying symptom-related terms for Google Trends searches.

The Google Trends search index data were extracted at the federative unit level, the highest geographic resolution available through the platform. The dataset covers the period from 2019 to 2024, as a five-year range is the maximum timespan for which the API provides weekly search index values. Although longer periods can be requested, the API returns data at a monthly resolution, which is not aligned with the epidemiological week format used in our official disease records. Additionally, because Google Trends normalizes the index within each extraction, combining data from different time windows can result in scale inconsistencies. To address this issue, values must be rescaled to ensure comparability. We present a normalization method for aligning time series across extractions, as detailed in **Usage Notes** section.

Alongside the search index for each query topic, we extracted additional data from the Related Topics and Related Queries sections available through the Google Trends platform. These features provide insights into user behavior by identifying topics and queries commonly searched in association with the original term. Related Topics refer to broader subjects linked to the query, while Related Queries capture specific search phrases entered by users. Both are ranked using two metrics: "Top" and "Rising". Top indicates the most frequently searched items on a relative scale (where 100 represents the highest popularity). Rising highlights items with the greatest increase in search frequency over the preceding period. For each disease (dengue, chikungunya, influenza, and COVID-19) we collected Related Topics and Related Queries sorted by the Top metric, on a monthly basis, for each federative unit and for Brazil as a whole, covering the period from 2020 to 2024.

Data processing. *Official public sources.* For all diseases, we collected the report date, notification date, symptom onset date, federative unit of notification, and clinical symptoms. Although the data extracted from public databases are individual-level records, by the end of the processing workflow, we present the data as aggregated counts of cases and symptoms, grouped by epidemiological week and federative unit. After extracting the datasets, we selected variables of interest based on their availability within each dataset and across the years covered. For the SINAN dataset, we calculated the difference in days between the notification date and the symptom onset date. Records in which this difference exceeded 180 days were excluded to reduce distortions caused by errors, such as data entry errors or incorrect patient date of birth entries in the symptom onset field. Both the symptom onset and notification dates were converted to their corresponding epidemiological week start dates. An epidemiological week is a standardized time unit used in public health surveillance, typically defined as a seven-day period that starts on Sunday and ends on Saturday, facilitating consistent reporting and analysis of disease trends across different regions' timeframes.

Symptom data were included starting in 2013 for dengue and in 2017 for chikungunya, reflecting the addition of these variables in notification forms only from those years onward. For symptom-related variables, we applied a standardization rule in which all values except for "1" (indicating "Yes") were converted to missing values (NA). Finally, the data were aggregated by epidemiological week (based on symptom onset, data entry, and notification dates), federative unit, and final case classification. For each combination, we calculated the total number of dengue and chikungunya cases, as well as the number of cases in which each symptom was reported. Note that each case may or may not include symptom information; thus, symptom counts reflect the number of cases reporting each symptom within each group.

For dengue, we included all cases reported in the SINAN-DENGUE system, excluding those classified as "discarded," i.e., cases with a final classification different from code "5". For chikungunya, we included all cases reported in the SINAN-CHIK database that were confirmed as chikungunya, identified by a final classification code equal to "13". For the SIVEP data, cases with a final classification of "2" (SARI due to another respiratory virus) or "3" (SARI due to another etiological agent) were grouped into a single category, while values of "9" were treated as "Not Available" (NA). For COVID-19 case counts, we considered SIVEP records with a final classification of "5" (SARI due to COVID-19), or those that tested positive via PCR or antigen tests for SARS-CoV-2.

Topic type	Topic	Topic (Portuguese)	Freebase ID	Alternative and colloquial terms
Disease	Chikungunya	Chikungunya	/m/01__7l	**
Disease	COVID-19	COVID-19	*	COVID-19, covid-19, covid, sarscov2, covid 19, covid19
Disease	Dengue	Dengue	/m/09wsg	**
Disease	Influenza/flu	Influenza/gripe	/m/0cycc	**
Symptom	Alteration of smell or taste	Alteração de olfato ou paladar	/m/01d3gn and /m/04czcv_	boca amarga
Symptom	Arthralgia	Artralgia	/m/021hck	dor na junta, dor no corpo, dor nas articulações, dor na munheca, dor articular, quebradeira, derrubada, dor nos ossos
Symptom	Arthritis	Artrite	/m/0t1t	inchaço nas juntas, inchaço nas dobras, inchaço, inchaço articular
Symptom	Chest pain	Dor no peito	/m/02np4v	aflição no peito, pressão no peito, aperto no peito
Symptom	Chills	Calafrio	*	tremedeira, arrepio, frio
Symptom	Cognitive dysfunction	Disfunção cognitiva	*	abobado,lesado, tontura, leseira, confuso, pensamento lento, abestado, leso, abobalhado, disfunção cognitiva
Symptom	Cough	Tosse	/m/01b_21	tosse, pigarro
Symptom	Diarrhea	Diarreia	/m/0f3kl	caganeira, dor de barriga, desarranjo, intestino solto, piriri
Symptom	Dyspnea	Dispneia	/m/01cdt5	falta de ar, dificuldade de respirar, sem ar
Symptom	Fatigue	Fadiga	/m/01j6t0	cansaço, moleza, fraqueza, corpo mole, piema
Symptom	Fever	Febre	/m/0cjf0	corpo esquentando, suadeira, corpo quente
Symptom	Headache	Cefaleia	/m/0j5fv	dor de cabeça, enxaqueca
Symptom	Loss of smell	Perda de olfato	/m/0m7pl	**
Symptom	Loss of smell or taste	Perda de olfato ou paladar	/m/0m7pl and /m/05sfr2	não sentir gosto e cheiro
Symptom	Loss of taste	Perda do paladar	/m/05sfr2	**
Symptom	Muscle stiffness	Rigidez muscular	*	braços ou pernas endurecidas, braço endurecido, braços endurecidos, perna endurecida, pernas endurecidas, juntas duras, corpo travado, entrevado, dificuldade em movimentar
Symptom	Myalgia	Mialgia	/m/013677	dor no corpo, dor muscular, dor nas pernas
Symptom	Nausea	Náusea	/m/0gxb2	estômago embrulhado, enjôo, estômago ruim, tonteira, vontade de vomitar, sensação de cabeça girando
Symptom	Retro-orbital pain	Dor retro-orbital	*	dor ocular, dor atrás dos olhos, dor no olho, vista doendo, olhos ardidos
Symptom	Skin rash	Erupção cutânea	*	calombos na pele, mancha na pele, caroço, coceira, perebas, brotoeja, ferida na pele, irritação, bolha no corpo, pipoco de pele, furunco, carnegão, pintas, vermelhidão, erupção cutânea
Symptom	Sore throat	Dor de garganta	/m/0b76bty	garganta doída
Symptom	Vomiting	Vômito	/m/012qjw	**

Table 4. Detailed list of topics, search terms, Freebase ID codes, and their variations used in this study. *FreebaseID code does not exist for this topic. **Alternative terms were not included for this topic.

For influenza case counts, we considered records with a final classification of "2," or those that tested positive via PCR or antigen tests for influenza. In instances where SARI cases tested positive for both COVID-19 and influenza, a new final classification value of "0" was assigned to indicate dual infection. Similar to the SINAN dataset, for symptom variables, any values other than "1" (indicating "Yes") were converted to NA. This process resulted in the creation of three intermediate tables — **SINAN_dengue_cases, SINAN_chik_cases**, and **SIVEP_cases** — all of which are available in our data repository. We then combined the case counts from both data sources (SINAN and SIVEP) into a single final dataset that provides disease case counts aggregated by: (i) epidemiological week of symptom onset, and (ii) federative unit. This final aggregated table is referred to as the **Arbo_SARI_disease_table** in the data repository.

For each epidemiological week and federative unit, we counted the number of cases for the following categories: dengue, chikungunya, COVID-19, influenza, SARI caused by other respiratory viruses, SARI cases with no etiological agent information available (i.e., final classification or PCR and antigen test results unavailable), and the total number of SARI cases. The criteria for classifying cases in each disease category are summarized in Table 5. For dengue and chikungunya cases, we included both confirmed and suspected cases, excluding records with a final classification of "5" (discarded) or NA (Not Available). In the case of SARI, when a single record tested positive for both influenza and COVID-19, it was included in the time series counts for each disease but counted only once in the "total SARI cases" series. For diseases with data outside the available time frame in the respective dataset, the value is recorded as NA.

Web search data. After extracting interest indexes for all search terms from Google Trends, we organized the data aggregated by week and federative unit into a table, which is available in our data repository under the name **GoogleTrends_search**.

To identify patterns related to public interest in symptoms, we cross-referenced the Relate Topics associated from each disease (dengue, chikungunya, influenza, and COVID-19) with a predefined list of key symptoms for each disease, as outlined in Table 3. We also included topics that explicitly contained the word "symptom" (in Portuguese, "Sintoma"). Based on this approach, we created a binary variable, *key_symptom*, which indicates

Database	Disease	Disease classification criteria	Criteria description
SINAN	Dengue	CLASSI_FIN ≠ 5	All dengue cases from SINAN-DENGUE, except 'discarded' records
SINAN	Chikungunya	CLASSI_FIN = 13	All chikungunya cases from SINAN-CHIK classified as chikungunya
SIVEP	COVID-19	CLASSI_FIN = 5 OR PCR_SARS2 = 1 OR AN_SARS2 = 1	SARI cases from SIVEP-Gripe classified as COVID-19 or positive for PCR or antigen test for COVID-19
SIVEP	Influenza	CLASSI_FIN = 1 OR POS_PCRFLU = 1 OR POS_AN_FLU = 1	SARI cases from SIVEP-Gripe classified as influenza or positive for PCR or antigen test for influenza
SIVEP	Other respiratory virus-induced SARI	CLASSI_FIN = 2 OR CLASSI_FIN = 3	SARI cases from SIVEP-Gripe with an unidentified etiological agent or one that is neither influenza nor COVID-19
SIVEP	SARI with unknown etiological agent	CLASSI_FIN = NA	SARI cases from SIVEP-Gripe without Final Classification, that is neither influenza nor COVID-19
SIVEP	Total SARI Cases	All CLASSI_FIN categories	All SARI cases from SIVEP-Gripe

Table 5. Criteria for classifying cases in each disease category.

whether a given topic in the Related Topic database was related to the characteristic symptom of the corresponding disease. This information is stored in the GoogleTrends_related_topic table in the data repository. The corresponding data on *Related Queries* are available in the GoogleTrends_related_query table. A schematic overview of the data collection and integration process is presented in Fig. 2.

Data Records

The dataset presented in this work is publicly available via Zenodo[52] and contains detailed records and metadata organized in a clear structure for ease of access and analysis. The data are structured into separate files, each focusing on specific variables and classifications relevant to the research.

Official public dataset.

To facilitate intermediate data processing, specific datasets were created for dengue, chikungunya, and SARI cases. These include **SINAN_dengue_cases** (dengue), **SINAN_chik_cases** (chikungunya), and **SIVEP_cases** (SARI). Each dataset contains key variables such as the start date of the epidemiological week for the registration, notification, and symptom onset dates; federative unit abbreviation; final case classification; case count; and presence of symptoms count. Metadata tables describing these intermediate datasets—including epidemiological variables, clinical symptoms, and classification criteria—are presented in Tables 6, 7. While case counts for dengue and chikungunya follow a straightforward structure, for **SIVEP_cases**, case counts are provided both according to the original final classification and after category regrouping, as detailed in the Data Processing section (see Table 4).

Finally, we present the **Arbo_SARI_disease_table**, which consolidates case counts for each disease—including arboviral diseases from SINAN and SARI cases from SIVEP—aggregated by epidemiological week and federative unit, as well as for Brazil as a whole. Due to varying time coverage across diseases, missing data for specific diseases on certain dates are represented as *NA* (Not Available), allowing for a consistent dataset structure while transparently reflecting temporal gaps in data availability. The dataset is cited by its corresponding repository name and includes detailed metadata to ensure proper reuse. A metadata table focusing on the case count data aggregated by the epidemiological week of symptom onset and grouped by federative unit, covering all diseases and search interests for distinct topics, is presented in Table 8.

Web search dataset.

For the web search data, we provide three distinct tables. The first, **GoogleTrends_search**, contains normalized search interest scores for terms related to diseases and symptoms across Brazilian federative units. The second, **GoogleTrends_related_topic**, includes data on the related topics associated with disease-related searches, while the third, **GoogleTrends_related_query**, presents the most frequently related queries by users regarding those diseases. Metadata for the **GoogleTrends_search** table, detailing search interest by federative unit, epidemiological week, and search term, is provided in Table 9. Metadata for **GoogleTrends_related_query** and **GoogleTrends_related_topic** are presented in Tables 10, 11, respectively.

To further illustrate the data made available, we present incidence trends for each disease and their associated symptoms—based on both official public health data and Google Trends search activity—across all federative units, by epidemiological week, in the Supplementary Material. This material also includes monthly trends of Related Topics associated with dengue. These data offer a descriptive overview of public search behavior related to the disease, highlighting the symptom-related aspects that generate the most user interest. These visualizations are intended to support potential correlation analyses between online search behavior and official case reports, offering insights into how digital surveillance can complement and enhance traditional epidemiological monitoring.

Technical Validation

Official public data validation.

Considering that the SINAN and SIVEP notification systems have changed certain variable codings over time, we reviewed original forms and official government documentation, including data dictionaries, to ensure data harmonization and consistency. Descriptive plots were created to assess the consistency and coherence of the collected data, as exemplified in Supplementary Material. We examined the formats of the extracted variables (i.e., numerical, string, categorical) and confirmed whether the dates were in the correct format and fell within logical ranges. Values with data entry errors or missing information were removed from the dataset. For instance, we noted that the notification data for dengue and chikungunya cases in the state

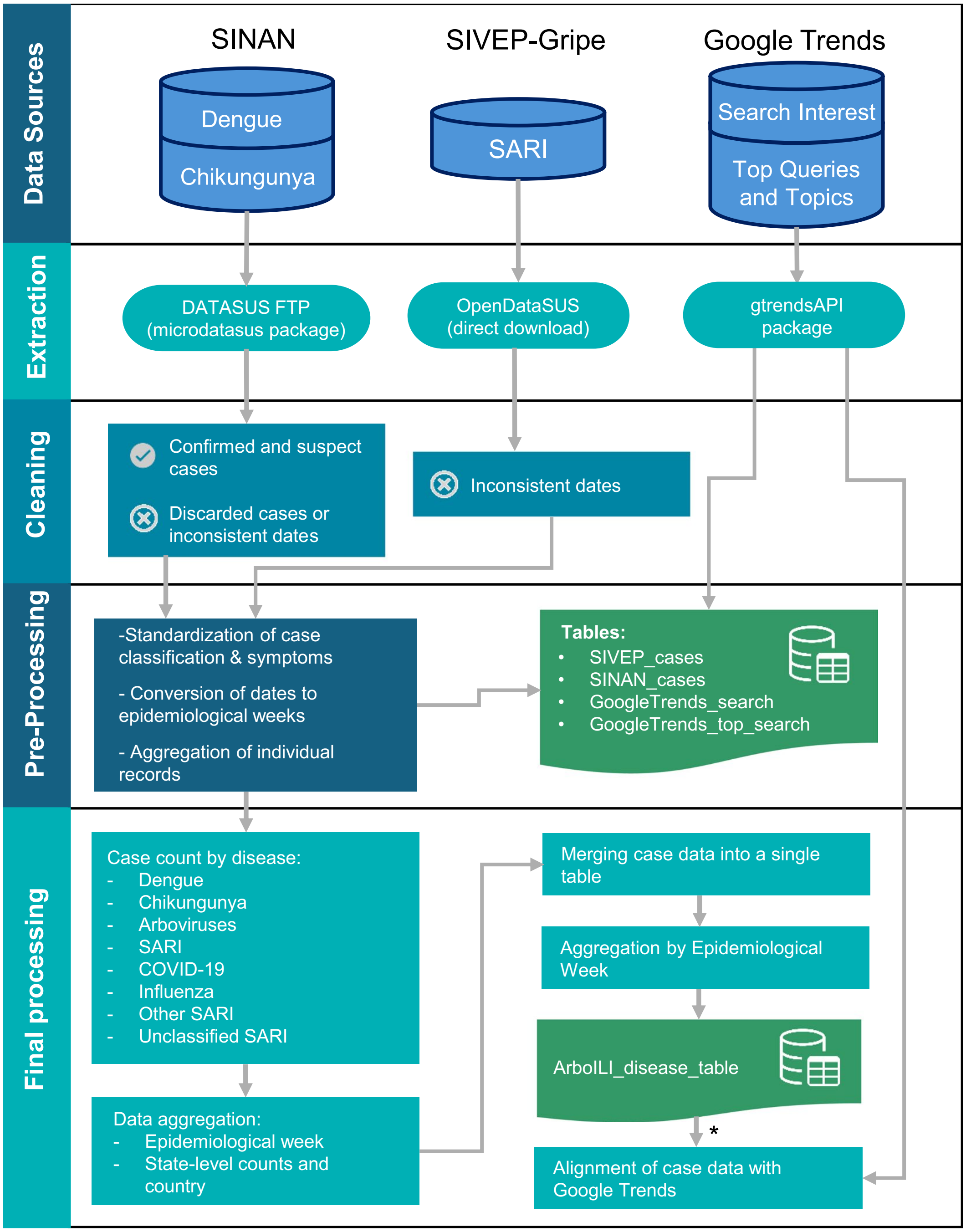

Fig. 2 Flowchart illustrating the data processing steps, from the extraction of original data sources to the preparation of the data tables. *The alignment step indicates the process of integrating the Arbo_SARI_disease_table with Google Trends search data (GoogleTrends_search table).

of Espírito Santo (ES) stopped being reported in SINAN starting in 2020, as they were transferred to a dedicated system called "Sistema de Informação em Saúde e-SUS Vigilância em Saúde (VS)"[53].

We also calculated the proportions of final classifications in the SINAN database by notification year, specifically for dengue and chikungunya cases. This analysis highlights changes in the distribution of dengue case classifications over time, showing how the proportions of "Classic Dengue," "Dengue with complications," "Dengue Hemorrhagic Fever," and other categories have varied throughout the year, as demonstrated in Fig. 3. The shifts in classification categories reflect evolving trends in case severity and reporting, providing valuable insights into the progression of the epidemic across different periods. Additionally, it is important to note that between 2014 and 2016, some chikungunya records in the database were misclassified under categories related to dengue. After 2016, the final classification for chikungunya cases was simplified to only three categories: "Discarded," "Chikungunya," or "Not Available," as shown in Fig. 4. Since the analysis period for comparing data with Google

Variable	Description	Type	Values or Format
ew_recorded	First day of the epidemiological week of data entry	Date	YYYY-MM-DD
ew_notification	First day of the epidemiological week of notification	Date	YYYY-MM-DD
ew_symptom_onset	First day of the epidemiological week of symptom onset	Date	YYYY-MM-DD
final_classification	Final case classification	Categorical	1-Classic Dengue; 2-Dengue with complications; 3-Dengue Hemorrhagic Fever; 4-Dengue Shock Syndrome; 5-Discarded; 6-Undefined code (entry error); 8-Inconclusive; 10-Dengue; 11-Dengue with Warning Signs; 12-Severe Dengue; 13-Chikungunya
state_abbrev	Federative unit of notification	Categorical	State abbreviation
case_count	Number of cases	Integer	≥ 0
fever	Symptom: fever	Binary	1 = Yes, 0 = No or unknown
myalgia	Symptom: muscle pain	Binary	1 = Yes, 0 = No or unknown
headache	Symptom: headache	Binary	1 = Yes, 0 = No or unknown
rash	Symptom: rash	Binary	1 = Yes, 0 = No or unknown
vomiting	Symptom: vomiting	Binary	1 = Yes, 0 = No or unknown
nausea	Symptom: nausea	Binary	1 = Yes, 0 = No or unknown
back_pain	Symptom: back pain	Binary	1 = Yes, 0 = No or unknown
conjunctivitis	Symptom: conjunctivitis	Binary	1 = Yes, 0 = No or unknown
arthritis	Symptom: arthritis	Binary	1 = Yes, 0 = No or unknown
arthralgia	Symptom: joint pain	Binary	1 = Yes, 0 = No or unknown
petechiae	Symptom: petechiae	Binary	1 = Yes, 0 = No or unknown
leukopenia	Symptom: leukopenia	Binary	1 = Yes, 0 = No or unknown
tourniquet_test	Tourniquet test result	Binary	1 = Positive, 0 = No or unknown
retro-orbital_pain	Symptom: retro-orbital pain	Binary	1 = Yes, 0 = No or unknown

Table 6. Metadata for the intermediate processed dataset (SINAN_*_cases) includes case-level records of dengue and chikungunya reported in the SINAN system. The table offers definitions and descriptions of the epidemiological variables, clinical symptoms, and classification criteria included in the dataset.

Trends focuses on the past five years, these early registration issues do not affect potential comparisons between disease notifications and Google Trends data.

Lastly, we compared the number of dengue and chikungunya cases extracted from SINAN, processed using the previously described methodology, with the publicly available case numbers from TabNet (https://datasus.saude.gov.br/informacoes-de-saude-tabnet/). TabNet is a public health database managed by the Brazilian Ministry of Health that provides aggregated epidemiological data. We compiled the case numbers by year of notification and federative unit. For each federative unit and year, we calculated both the absolute difference in case numbers and the relative difference compared to the TabNet data. It is worth noting that for the year 2024, the relative difference exhibits a slight increase, which may be attributed to delays in data updates within the TabNet system. This discrepancy could also be partially explained by the unusually high incidence of dengue cases in that year[54], potentially impacting the timeliness of data processing and reporting. For chikungunya, we considered three scenarios: total cases from the system, cases classified with Final Classification equal to 13 (corresponding to confirmed chikungunya cases), and all cases except those with Final Classification equal to 5 (i.e., discarded cases). The comparative analysis of the relative differences for dengue and chikungunya is presented in Figs. 5, 6, respectively.

Web search data validation. The data extracted via the Google Trends API was compared with searches conducted using the online tool to validate whether the extraction results aligned with the data provided by the website. This comparison was carried out for various search terms and states, considering a five-year time frame.

Statistical validation. Moreover, the time series data for both disease notification cases and Google Trends search terms were visually examined using dynamically generated graphs to ensure that the representations aligned with expected trends.

Usage Notes

Since the final databases are available for download as CSV files, researchers can analyze the combined dataset using R, Python, or other statistical tools for time series analysis, regression, or spatial modeling. The script used for data extraction and processing is publicly accessible at: https://github.com/arboiliproject/Arboili_datapaper.

When working with Google Trends data extracted from different periods, it is necessary to align the data curves due to the normalization applied within each extraction. As each dataset is scaled independently to range from 0 to 100, combining information from multiple time intervals—such as integrating recent data with

Variable	Description	Type	Values or Format
ew_recorded	First day of the epidemiological week of data entry	Date	YYYY-MM-DD
ew_notification	First day of the epidemiological week of notification	Date	YYYY-MM-DD
ew_symptom_onset	First day of the epidemiological week of symptom onset	Date	YYYY-MM-DD
final_classification_original	Original classification categories from SINAN	Categorical	1-SARI due to influenza; 2-SARI due to another respiratory virus; 3-SARI due to another etiological agent (specified in the CLASSI_OUT column); 4-SARI not specified; 5-SARI due to COVID-19;
state_abbrev	Federative unit of notification	Categorical	State abbreviation
final_classification_new	Regrouped classification categories	Categorical	0-SARI with COVID-19 and influenza detected simultaneously; 1-SARI due to influenza; 2-SARI due to another respiratory virus; 4-SARI not specified; 5-SARI due to COVID-19;
case_count	Number of cases	Integer	≥0
fever	Symptom: fever	Binary	1 = Yes 0 = No or unknown
cough	Symptom: cough	Binary	1 = Yes 0 = No or unknown
sore_throat	Symptom: sore throat	Binary	1 = Yes 0 = No or unknown
breath_shortness	Symptom: shortness of breath	Binary	1 = Yes 0 = No or unknown
resp_distress	Symptom: respiratory distress	Binary	1 = Yes 0 = No or unknown
diarrhea	Symptom: diarrhea	Binary	1 = Yes 0 = No or unknown
vomiting	Symptom: vomiting	Binary	1 = Yes 0 = No or unknown
smell_loss	Symptom: loss of smell	Binary	1 = Yes 0 = No or unknown
taste_loss	Symptom: loss of taste	Binary	1 = Yes 0 = No or unknown
ab_pain	Symptom: abdominal pain	Binary	1 = Yes 0 = No or unknown
fatigue	Symptom: fatigue	Binary	1 = Yes 0 = No or unknown

Table 7. Metadata for the intermediate processed dataset (SIVEP_cases) contains case-level records of Severe Acute Respiratory Infection (SARI) reported in the SIVEP system. The table provides definitions and descriptions for the epidemiological variables, clinical symptoms, and classification criteria included in the dataset.

Variable	Description	Type	Values
ew_symptom_onset	Epidemiological week of symptom onset	Date	YYYY-MM-DD
location	Geographic region for case count	Categorical	State/country abbreviation
dengue_cases	Number of dengue cases	Integer	≥0
chik_cases	Number of chikungunya cases	Integer	≥0
arbo_cases	Number of Arboviral cases (dengue and chikungunya)	Integer	≥0
sari_cases	Number of Severe Acute Respiratory Syndrome (SARI) cases	Integer	≥0
covid_cases	Number of COVID-19 cases	Integer	≥0
flu_cases	Number of influenza cases	Integer	≥0
other_sari_cases	Number of SARI cases from other respiratory viruses	Integer	≥0
na_sari_cases	Number of SARI cases with final classification not available	Integer	≥0

Table 8. Metadata for the final dataset of reported cases by epidemiological week (Arbo_SARI_disease_table). This dataset includes the number of cases for various diseases, grouped by federative unit.

older series or extending analyses beyond five years—requires an adjustment to reconcile the differing scales. A useful approach involves "aligning" the curves from two overlapping periods, allowing for the calculation of a normalization factor.

As an example, two extractions were performed using the Freebase ID for Dengue (*/m/09wsg*) for the state of São Paulo. The first extraction covers the period from June 2016 to June 2020, while the second spans June 2019 to June 2024, resulting in a two-year overlap (Fig. 7A). The code provided in our repository automates the alignment process through a series of transformations. First, it reshapes the datasets, organizing search interest values into distinct columns for each extraction, ensuring that each row corresponds to a unique date. It then removes any incomplete rows, retaining only fully available data. Next, the code calculates the ratio between the overlapping interest values to derive a set of scaling factors, and computes their average to serve as a normalization factor. This factor is applied to adjust one of the datasets, bringing both onto a comparable scale. Finally, the normalization factor is stored for potential use in future adjustments. This method allows for more accurate comparisons between datasets spanning different extraction periods (Fig. 7).

Variable	Description	Type	Values
date	Start date of the epidemiological week	Date	YYYY-MM-DD
location	Geographic region of query	Categorical	Location (State/country abbreviation)
topic	Search term used in Google Trends	Categorical	Search term (e.g. "Dengue", "Chikungunya", "Influenza", and "COVID-19)
value	Google Trends index value	Numeric	0–100 (trend index value)

Table 9. Metadata for the final Google Trends dataset (GT_dataset), used to track public health trends based on search behavior. The table includes variables such as the start date of the epidemiological week, federative unit, search terms queried in Google Trends, and the corresponding trend index values.

Variable	Description	Type	Values
topic_title	Topic name associated with disease search	String	Topic name (e.g. Fever, Headache, Symptom)
topic_id	Topic FreeBase ID	String	Code (e.g. "*/m/01b_06*")
date	Month and year of related topic metric	Date	YYYY-MM
location	Geographic region of query	Categorical	Location (State/country abbreviation)
value	Google Trends index value	Numeric	0–100 (trend index value)
disease	Disease name	Categorical	Search term (e.g. "Dengue", "Chikungunya", "Influenza", and "COVID-19)
key_symptom	Topic related to the disease key symptoms	Logical	TRUE or FALSE

Table 10. Metadata for the final Google Trends Related Topics dataset (GoogleTrends_related_topic), used to track public interest in topics associated with dengue, chikungunya, influenza, and COVID-19. The table includes variables such as the epidemiological week, geographic location, topic name and ID, trend index values, and a binary indicator for symptom-related topics.

Variable	Description	Type	Values
top_search	Queries related to the disease search	String	Topic name (e.g. "covid symptoms", "spanish flu")
date	Month and year of related topic metric	Date	YYYY-MM
location	Geographic region of query	Categorical	Location (State/country abbreviation)
value	Google Trends index value	Numeric	0–100 (trend index value)
disease	Disease name	Categorical	Search term (e.g. "Dengue", "Chikungunya", "Influenza", and "COVID-19)

Table 11. Metadata for the final Google Trends Related Queries dataset (GoogleTrends_related_query) captures the most frequently searched queries related to dengue, chikungunya, influenza, and COVID-19. The table includes information on the epidemiological week, geographic location, query terms, and corresponding trend index values.

Finally, to compare disease databases aggregated by epidemiological week, the Google Trends search interest data must be aligned to the same time scale, specifically by weeks. The Google Trends API, accessed through the gtrendsAPI package, allows users to specify a start and end date for data extraction, but it does not allow for direct specification of the frequency. Depending on the time period, the data returned may be aggregated at different time scales, such as daily, weekly, or monthly. The maximum period for a single extraction with weekly aggregation is up to five years. Periods longer than five years will return results on a monthly scale, while periods shorter than one month may return data on a daily or hourly scale. When data is aggregated by week, the reference date is the first day of the week (Sunday), which aligns with the start of the epidemiological week, enabling a direct comparison between search volume for that week and the number of reported cases.

For comparison between official disease notifications and Google search interest data, both the case counts and the interest index can be aligned by the date corresponding to the first day of each epidemiological week. Figure 8 illustrates Google Trends search interest for the term "Dengue" in Brazil over the past five years, alongside the official weekly dengue case counts reported during the same period. To facilitate visual comparison, the number of cases was rescaled to match the 0–100 range of the Google Trends interest index.

Data scope and limitations.

Public disease notification. The use of notifiable disease data is subject to limitations in both quality and coverage, largely due to underreporting. Many cases may go unregistered as a result of inefficiencies in health information systems and limited diagnostic capacity. Furthermore, challenges in distinguishing diseases with overlapping symptoms, particularly among arboviruses and respiratory infections, also contribute to underreporting. These limitations are especially pronounced in low-resource regions of Brazil, where healthcare infrastructure and reporting systems are often underdeveloped or inconsistently applied. Regional disparities in reporting practices further affect data reliability. While some federative units maintain comprehensive and timely records, others report less consistently, introducing spatial heterogeneity that can bias analyses and reduce the generalizability of findings at the national level.

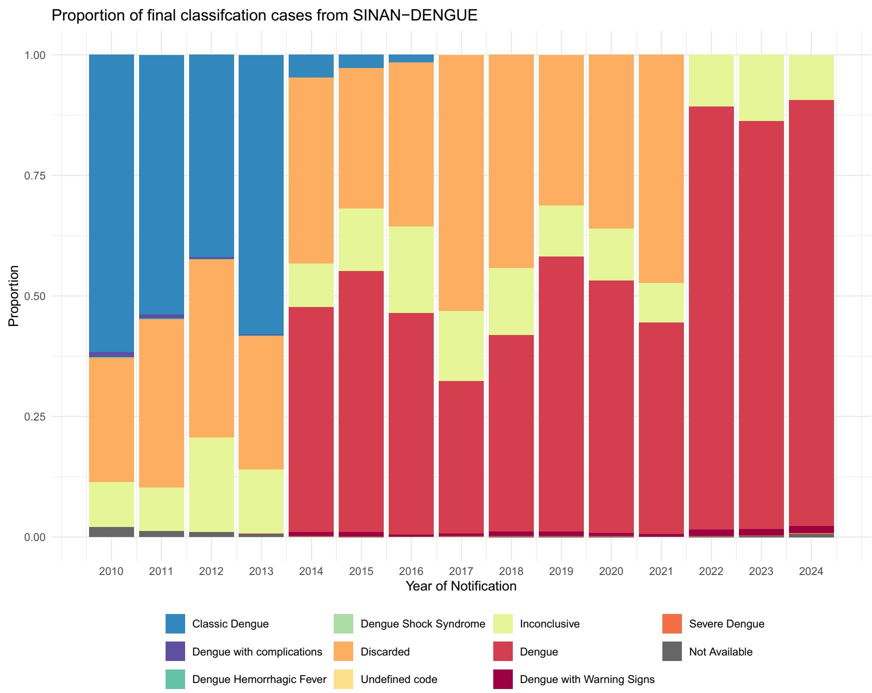

Fig. 3 Proportion of dengue cases by final classification and year of notification (SINAN-Dengue).

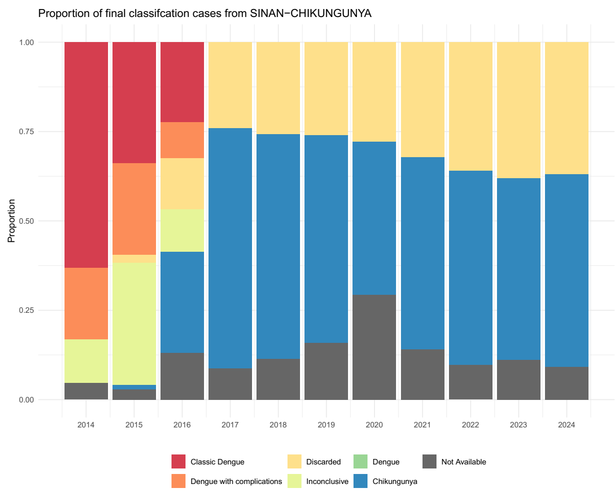

Fig. 4 Proportion of chikungunya cases by final classification and year of notification (SINAN-Chikungunya).

Recent epidemiological data are subject to inherent delays between the onset of illness, case notification, data entry, and public dissemination. As a result, the most recent weeks in the dataset are always incomplete due to reporting lags. To account for this temporal delay, analyses should ideally incorporate statistical methods such as nowcasting, which allow for the adjustment of case estimates for the most recent dates. Additionally, recently reported arboviral cases may be reclassified or discarded after further epidemiological investigation, which can affect the accuracy of most recent reporting periods.

For respiratory viruses, the available data primarily reflect severe cases requiring hospitalization, as recorded under the Severe Acute Respiratory Infection (SARI) surveillance system. This focus limits the ability to capture the broader spectrum of illness, particularly mild or moderate cases that do not result in hospitalization or laboratory confirmation. Even when individuals seek medical care, diagnostic testing to identify the specific infectious agent is not always conducted, leading to gaps in surveillance and potential underestimation of true disease burden.

Furthermore, as previously noted, dengue and chikungunya cases in Espírito Santo have not been reported through SINAN since 2020. As a result, data from this state cannot be used to estimate dengue incidence in more recent years.

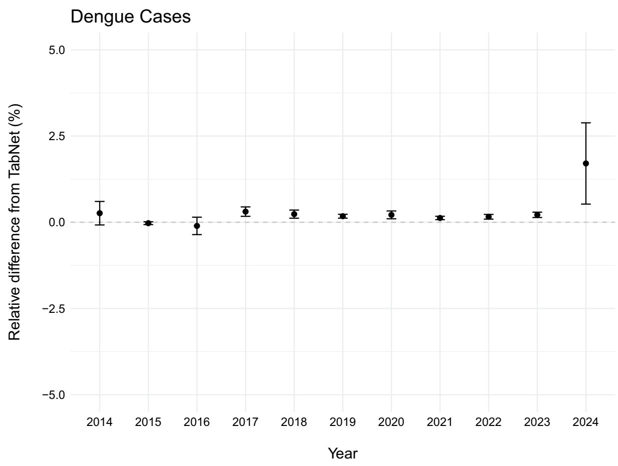

Fig. 5 Relative difference in the number of dengue cases between SINAN-processed data and TabNet by federative unit and year of notification. The relative difference is calculated with respect to the TabNet case numbers.

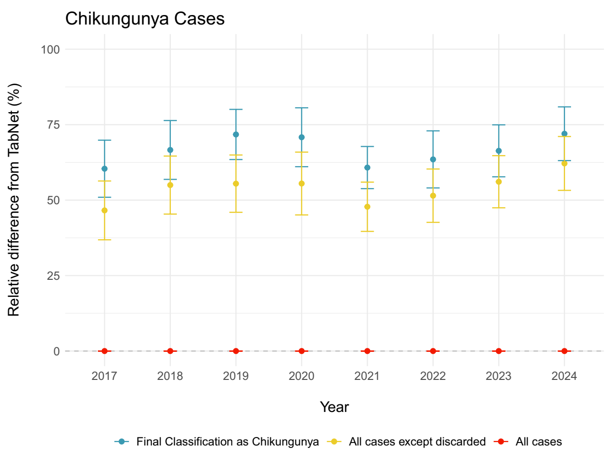

Fig. 6 Relative difference in the number of chikungunya cases between SINAN-processed data and TabNet, by federative unit and year of notification. Three case definitions were considered for chikungunya: (i) total cases from the system, (ii) cases with Final Classification = 13 (confirmed chikungunya cases), and (iii) all cases except those classified as Final Classification = 5 (discarded cases). The relative difference is calculated with respect to the TabNet case numbers.

Online search data. Search data—such as that available through Google Trends—poses specific challenges when applied to public health surveillance. One key limitation is its spatial granularity: the data are aggregated at the federative unit level, which masks local variations and dynamics essential for understanding disease patterns at the municipal level. A further limitation is the inherent bias associated with internet access. In many regions, particularly low-income or rural areas, internet connectivity remains limited or inconsistent. As a result, these populations are often underrepresented in search data, potentially skewing analyses toward urban and wealthier areas with higher internet penetration.

Despite efforts to capture regional variation in search terminology, as shown in Table 4, the use of predefined search terms introduces potential cultural and linguistic biases. Different populations may use distinct expressions to describe the same symptoms or diseases, which can result in incomplete or skewed representations of search behavior. Moreover, fluctuations in search volume may be influenced by external factors such as seasonal trends or media coverage, which do not necessarily reflect actual disease prevalence. These effects may lead to false positives or misinterpretations in the analysis.

Data availability

The dataset is available in the Zenodo repository at https://doi.org/10.5281/zenodo.17102575.

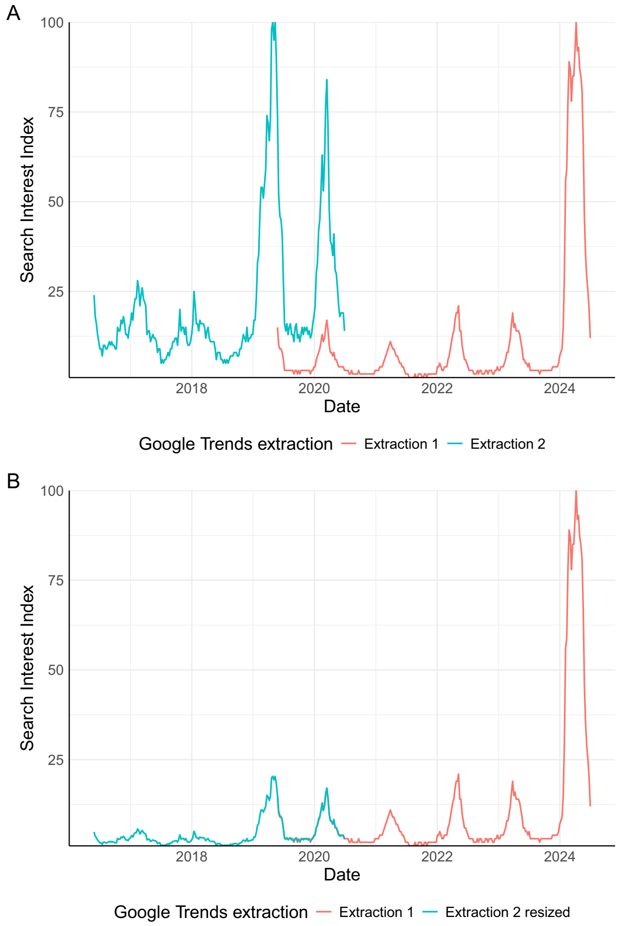

Fig. 7 Alignment of Google Trends search interest data extracted from different periods. (**A**) Overlapping search interest data for the term "Dengue" in the state of São Paulo, Brazil. Two extractions were performed: the first from June 2016 to June 2020, and the second from June 2019 to June 2024, resulting in a two-year overlap used for alignment. (**B**) Result of the alignment process after applying the normalization factor calculated from the overlapping period, allowing the two time series to be combined on a comparable scale.

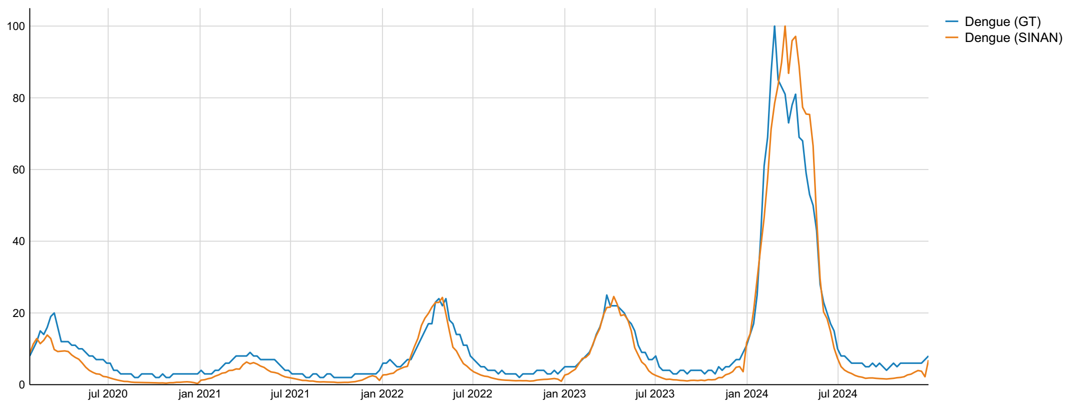

Fig. 8 Google Trends search trends for the term "Dengue" in Brazil over the past 5 years compared to the official dengue cases reported of dengue cases during the same period.

Code availability

All custom code used for data processing, cleaning, and validation is available at the GitHub repository: https://github.com/arboiliproject/Arboili_datapaper, including the scripts for data extraction from Google Trends and preprocessing of SINAN and SIVEP data.

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M.E.B.: Data Curation, Formal Analysis, Investigation, Methodology, Validation, Visualization, and Writing. A.R.M.A.: Conceptualization, Funding Acquisition, Investigation, Methodology, Project Administration, Resources, Supervision, Validation, and Writing. C.T.C.: Conceptualization, Funding Acquisition, Investigation, Methodology, Project Administration, Resources, Supervision, Validation, and Writing. D.M.B.O.: Methodology and Writing.

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